

Akintoye Ayomide Nelson, Senior Backend Engineer

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 7+ years of hands-on experience building and maintaining high-scale, event-driven backend systems using TypeScript + Node.js + NestJS in production fintech and transaction platforms. Specialized in PostgreSQL optimization, Redis, and resilient architectures with autoscaling, high availability, and fault tolerance. Production experience in AWS environments with Docker, CI/CD, and observability. Strong advocate of AI-assisted development (Claude + Cursor) and automated workflows.

TECHNICAL SKILLS

Core Backend: TypeScript, Node.js, NestJS (primary framework), Express, Kafka, Redis

Databases & ORM: PostgreSQL (Prisma, Drizzle – schema design, query optimization, indexing), MongoDB, MySQL

DevOps & Cloud: AWS, Docker, Kubernetes, CI/CD (GitHub Actions), multi-environment deployments

Observability & Reliability: Logs, metrics, tracing, performance profiling, bottleneck analysis

Other: Laravel (PHP), React (occasional frontend contributions), AI-assisted coding (Claude + Cursor)

WORK EXPERIENCE

PANDAR RESOURCES.

Mar 2026 - Current

Senior Backend Engineer

Lagos, Nigeria

- Built and optimized robust API integrations and data pipelines using Node.js, TypeScript, and NestJS with MongoDB, reducing average response times by 60% through efficient connection pooling and query optimization.
- Contributed to observability enhancements by implementing structured logging and metrics, enabling faster debugging of high-throughput services.

ODJTECH MULTIMEDIA.

Mar 2025 - Feb 2026

Senior Backend Engineer

Lagos, Nigeria

- Designed and maintained event-driven backend services using NestJS + TypeScript, integrating with high-volume data flows and applying business logic transformations to ensure reliable downstream processing.
- Optimized database operations and implemented caching strategies with Redis, significantly improving response times under concurrent load.
- Collaborated on system reliability improvements, including failure-mode handling and autoscaling patterns for production stability.

UNISPEND.

Mar 2025 - Feb 2026

Senior Backend Engineer

Lagos, Nigeria

- Designed and deployed secure, scalable backend systems with Node.js, TypeScript, and NestJS, integrating MongoDB for real-time student wallet top-ups, QR-code payments, and transaction tracking in a campus ecosystem.

- Optimized PostgreSQL/MongoDB queries and implemented JWT-based authentication with rate limiting, enhancing security and enabling analytics dashboards for revenue-sharing models.
- Built real-time notification flows and observability instrumentation (logs + metrics) to support high-concurrency transaction processing.

CHOPEXPRESS LTD.

Oct. 2024 - Jan 2025

Lead Backend Engineer

Lagos, Nigeria

- Architected high-performance, scalable RESTful APIs for a multi-tenant food delivery platform using Node.js, TypeScript, Express, NestJS patterns, and PostgreSQL (Prisma ORM), successfully handling high-concurrency rider, business, and order workflows.
- Refactored in-memory operations into optimized PostgreSQL queries with strategic indexing, reducing transaction lookup times by 50–60% and improving performance under peak load.
- Implemented resilient data pipelines and batch processing logic to support real-time order tracking at scale.

FASTPAY TECH

Jan. 2023 - Sept 2024

Fullstack Engineer

Abuja, Nigeria

- Developed scalable backend APIs using Node.js, Express, and PostgreSQL, handling high-volume payment transactions with AWS deployment for reliability and scalability.
- Designed and integrated a real-time webhook system for payment notifications, ensuring delivery guarantees and reducing manual reconciliation by 40%.
- Contributed to frontend payment interfaces while maintaining full ownership of backend performance and security.

PPELABS

Jun 2022 - Dec 2022

Backend Engineer

Houston, TX, USA

- Built secure RESTful endpoints in Laravel with JWT authentication, rate limiting, and input validation, reducing potential security vulnerabilities by 50%.
- Implemented webhook systems that improved payment processing reliability and partner integration efficiency..

TASKIMONY INC.

Jan 2020 - May 2022

Full Stack Engineer

Illinois, United States

- Led backend architecture and optimization for high-traffic web/mobile applications in an Agile environment, improving overall performance by 40% and reducing post-release defects by 25%

NTECH INFO SYSTEM.

Jun 2018 - Dec. 2019

Junior Backend Engineer

Akure, Ondo State

- Documented all existing RESTful API endpoints in Postman, improving team understanding and developer onboarding speed by 30-50% and decreasing support queries or integration issues by up to 35%.

PROJECTS

PLANTRIGHT AI [Link](#)

Full-Stack AI Platform for Reforestation

- Built and shipped a complete spatial AI platform using **Node.js + TypeScript, NestJS, Express, and PostgreSQL** with ArcGIS/QGIS integration for intelligent site selection, design, and monitoring of reforestation projects.
- Reduced manual workload for forestry teams, NGOs, and governments by 65–80% through automated ecological modeling

SELAR CLONE BACKEND [Link](#)

Scalable E-commerce Platform for Digital Creators (Africa-focused)

- Developed a robust, production-ready backend using Node.js, TypeScript, Express, NestJS patterns, and PostgreSQL (Prisma) to support digital product sales, memberships, bundles, and creator analytics.

EDUCATION

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE.	Aug 2023
Bachelor of Technology in Forestry and Wood Technology	Ondo State, Nigeria

AWARDS

Best Grandeza FUTA Tech Innovator, The Federal University of Technology, Akure.	May 2023
Best Agricultural Tech Innovator, National Association of Agricultural Students	May 2023
1st runner up Wema Bank Hackaholics Competition, Wema Bank Plc.	Feb 2022
Dean’s List award for First Class Honours, The Federal University of Technology, Akure	2019 - 2021

CERTIFICATIONS

Artificial Intelligence and Career Empowerment, University of Maryland	Jan 2026
Google Prompting Essentials Specialization, Google and Coursera	Jan 2026
AI for Business Innovation: From GenAI to POCs, Great Learning	Dec 2025
Net Zero 101, United Nations University Institute for the advance study of Sustainability	Nov 2024
Google Africa Developer Training Program, Google and Andela	June 2022
Next.js & Laravel 11 Build a Modern Full-Stack Application, Udemy	Jan 2021
NodeJS - The Complete Guide (MVC, REST APIs, GraphQL, Deno), Udemy	Jan 2021

PERSONAL SKILLS AND STRENGTHS

Leadership skills, excellent communication, Teamwork, decision making, presentation, Project management , creative thinking, collaborative skills and execution.